# MILITARY SPECIFICATION

# WELDING SET, ARC, INERT GAS SHIELDED

# PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1 SCOPE

1.1 Scope. This specification covers requirements for materiels and methods for cleaning, drying, preservation. packing, and marking of Welding Set, Arc, Inert Gas Shield, (SC 3431-97-CL-E02).

## 2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions, and Chemical Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing: With Appropriate Test Methods

(Copies of specifications and standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

## 3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete pack for first article inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Materials. Materials used in preservation and packing shall be as specified herein. Materials shall be new and free from defects and imperfections which may affect the serviceability of the completed package or pack (see 6.4 for disposability methods of materials).

3.3 Preservation. Preservation shall be level A or industrial, as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Methods of preservation. Methods of preservation specified herein by symbol shall conform to corresponding symbols of MIL-P-116.

3.3.1.2 Cushioning. Application of cushioning materials, for each method of preservation specified, shall be in accordance with the criteria and guidelines of MIL-P-116. Materials utilized shall conform to PPP-C-843, type II or PPP-C-1797. Other type cushioning materials may be used provided they are water resistant and provide protection equal to or better than the aforementioned materials.

3.3.1.3 Flow adapter, machine thread bushings, and hose glands. Items shall be individually packaged in accordance with submethod IC-1 criterion.

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3.3.1.4 Wire brushes, welding cable, ground clamps, hoses, pull gun, terminal lugs, and electrodes. Items of like description, in quantities required for each set, shall be packaged in accordance with method III criterion. Unit containers shall conform to PPP-B-636, class weather-resistant, style RSC or PPP-B-566. After closure, fiberboard boxes shall be waterproof sealed in accordance with the appendix to the box specification. Items packaged in paperboard boxes shall be intermediate packaged as specified in 3.3.1.8.

3.3.1.5 Regulator and wire guide kit. Components of the wire guide kit shall be individually wrapped, in quantities required for each kit, using barrier material conforming to MIL-B-121, grade A. Plastic cap-plugs conforming to MIL-C-5501 shall be placed over apertures of the regulator. The regulator and guide kit shall be individually packaged in accordance with submethod IC-2 criterion, utilizing a fiberboard box conforming to PPP-B-636, class domestic as the unit container. The regulator shall be blocked in a manner which will restrict the item from coming in contact with the inside surfaces of the container. Fiberboard conforming to PPP-F-320, class domestic shall be formed into pads, cells, or cut shapes to form the blocking media.

3.3.1.6 Secondary contactor, speed control, wire feeder, and water circulator. Items shall be individually packaged in accordance with method IIb criterion. The inner and outer containers shall conform to PPP-B-636, class domestic and class weather-resistant, respectively. The box closure shall be in accordance with the appendix to the box specification.

3.3.1.7 Argon cylinders. When specified in the contract or order, cylinders of argon shall be furnished with the welding set. Painted surfaces of the cylinders shall not be defaced with markings, except for the application of the appropriate precautionary marking. The valves' protective covers shall be firmly tightened.

3.3.1.8 Consolidated package. Components preserved by method IC and items packaged in paperboard boxes conforming to PPP-B-566 shall be intermediate packaged in a fiberboard box(es) conforming to PPP-B-636, class weather-resistant, style RSC. Cushioning material shall be provided to restrict movement of the contents. After closure, the box(es) shall be waterproof sealed as specified in 3.3.1.4.

3.3.2 Industrial. Welding set components shall be preserved and packaged in accordance with ASTM D3951-82.

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3.4 Packing. Packing shall be level A or B, or industrial, as specified (see 6.2).

3.4.1 Level A. Components comprising one complete set, except for cylinders of argon, shall be packed in a snug-fitting cleated plywood box conforming to PPP-B-601, overseas type. When specified (see 3.3.1.7), cylinders of argon shall be packed in a similar type container. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Box closure shall be in accordance with the appendix to the box specification.

3.4.2 Level B. Components comprising one complete set shall be packed as specified in 3.4.1, except shipping containers shall conform to domestic type.

3.4.3 Industrial. Each welding set shall be packed in accordance with ASTM D3951-82.

3.5 Marking.

3.5.1 Military. When military levels of protection are specified, exterior shipping containers shall be marked in accordance with MIL-STD-129 along with any special markings required by the contract or order.

3.5.2 Industrial. When commercial protection is specified, exterior containers shall be marked in accordance with ASTM D3951-82 along with any special markings required by the contract or order.

3.6 Workmanship. Standards of workmanship shall be such as to assure preservation, packing, and marking of a neat uniform general appearance in compliance with the requirements of section 3. Workmanship shall be of the best commercial practice. Defects and imperfections or omissions which would tend to impair the effectiveness of the completed pack or part thereof will not be permitted.

3.7 Performance.

* 3.7.1 Rough handling. When specified (see 6.2), the completed pack shall provide protection from rough handling sufficient to prevent damage to the packaged item(s) when the pack is subjected to the rough handling test of MIL-P-116. Following the tests, the packaged item(s) shall show no evidence of damage, and the exterior container shall not reveal any failure caused by improper application or design of interior cushioning, blocking, or bracing. All barriers, wrappings, interior containers, blocking, bracing and anchoring shall be intact and capable of providing the intended protection.



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* 3.7.2 Exposure. When specified (see 6.2), the completed pack shall provide protection from exposure to the elements sufficient to prevent damage to the packaged item(s) when the pack is subjected to the cyclic exposure test of MIL-P-116. Following the test, the tested package, when opened, shall show no evidence of moisture nor shall the packaged item(s) show corrosion.

## 4 QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When first article inspection is required by the contract or purchase order (see 3.1),, the contractor shall subject the first article pack to the applicable first article inspection and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), first article packages shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as specified in the first article inspection requirements of MIL-P-116. Failure of the first article to pass any of the inspections or tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to the production packs offered for acceptance under the contract or purchase order.

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4.2.2.1 Level A and B preservation and packing. Quality conformance inspection shall consist of the applicable quality conformance inspections and tests specified in MIL-P-116 and the examinations in the applicable container specification and Appendix B of MIL-STD-1186 to determine conformance to the requirements of this specification. When specified (see 6.2), the packs shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as specified in the quality conformance inspection requirements of MIL-P-116.

4.2.2.2 Industrial preservation and packing. Industrial preservation and packaging shall be examined to determine conformance with the requirements of ASTM D3951-82.

4.3 Sampling.

4.3.1 Level A and B preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in MIL-P-116 and Appendix B of MIL-STD-1186, respectively. Sampling for examination of the container shall be at level II and an AQL of 4.0. When specified (see 6.2), sampling for the rough handling and cyclic exposure test(s) (one or both) shall be at inspection level S-2 with an AWL of 4.0.

4.3.2 Industrial preservation and packing. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105 at inspection level S-4 with an AQL of 4.0.

4.3.3 Disposition of samples. All samples used for inspection and testing shall be reprocessed as necessary. They may, after reprocessing in accordance with the original method of preservation, be considered a part of the original lot. When the packaged item may have been damaged as a result of testing, the item shall be inspected as necessary to determine its acceptability.

5 PACKAGING

5.1 This section not applicable to this specification.

6 NOTES



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6.1 Intended use. It is intended that this specification be used for direct reference in contracts or orders for the preservation, packing, and marking of a Welding Set, Arc, Inert Gas Shielded (NSN 3431-00-731-4163).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Specify level A or B, or industrial preservation or packing (see 3.3 and 3.4).
(d) Rough handling and cyclic exposure tests, if required (see 3.x.1, 3.x.2, 4.2.1 and 4.2.2.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Disposability methods. Environmental Pollution Measures are contained in the materials specifications indicated herein. Refer to the materials specifications (or preparing activity) for recommended disposability methods.

Custodian:
Army - AL

Preparing activity:
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Project No. PACK-A280

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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
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| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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